



TB AUDIO PLUG-INS
Team Impulse Impact

USER MANUAL

TB Geometry Reverb

Ray-Traced Geometry Reverb

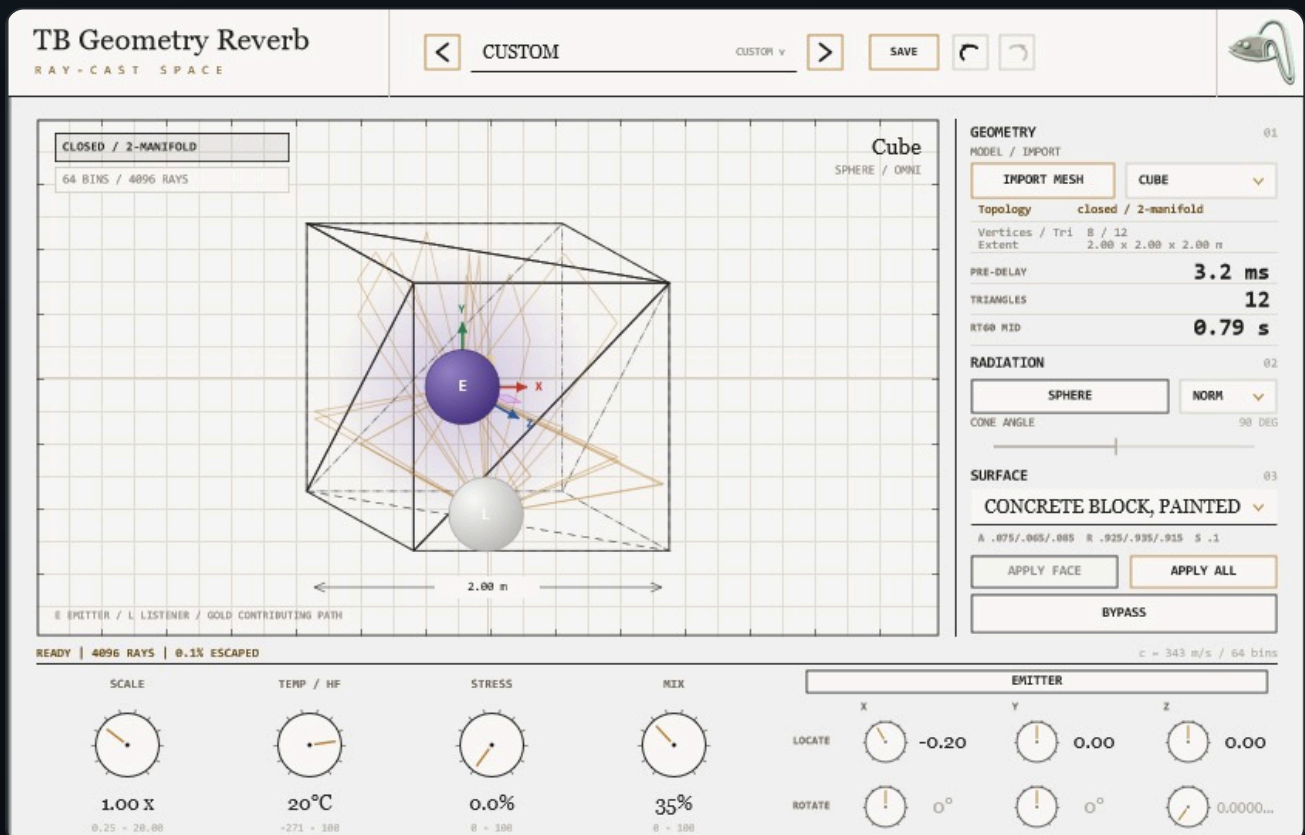
Builds a reverb from a 3D shape: rays travel through a closed mesh from an emitter to a listener, and its scale, materials, radiation, and positions shape the space you hear.

13 built-in models + OBJ/FBX import

Surface materials per face

Sphere or cone radiation

30 factory presets



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01 Overview

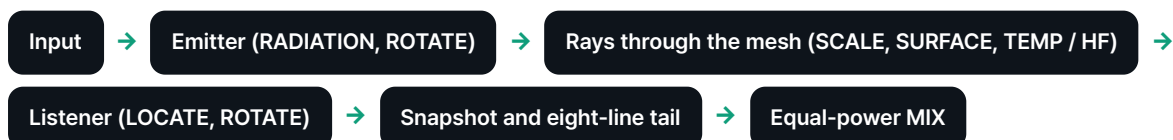
TB Geometry Reverb turns a 3D shape into a room. It traces sound rays through a closed mesh from an emitter (E) to a listener (L); the paths that reach the listener, the surface materials, the scale of the mesh, and the air temperature form an acoustic snapshot of the space, and an eight-line feedback delay network adds the dense tail.

Choose a built-in model or import your own OBJ or FBX mesh, assign materials to its surfaces, and place the emitter and listener inside it. **SCALE** sets the size, **TEMP / HF** the air, **RADIATION** how the emitter radiates, and **MIX** the equal-power dry/wet balance. The view shows the mesh with the ray paths that contribute to the sound.

Key features

- Ray-traced acoustic snapshot with 64 direction bins and three ray-quality levels
- 13 built-in models, from a cube to a trefoil knot and an auditorium, plus OBJ and FBX import
- Closed-mesh validation with live status and an escaped-ray readout
- Surface materials with absorption and scattering, per face or for all surfaces
- Sphere or cone radiation with a 10 to 180° cone
- Emitter and listener position and orientation in 3D
- SCALE from 0.25 to 20×, air temperature from −271 to 100 °C
- Calculated pre-delay, triangle count, and mid-frequency RT60
- 30 factory presets, user presets, undo, and redo

Signal flow



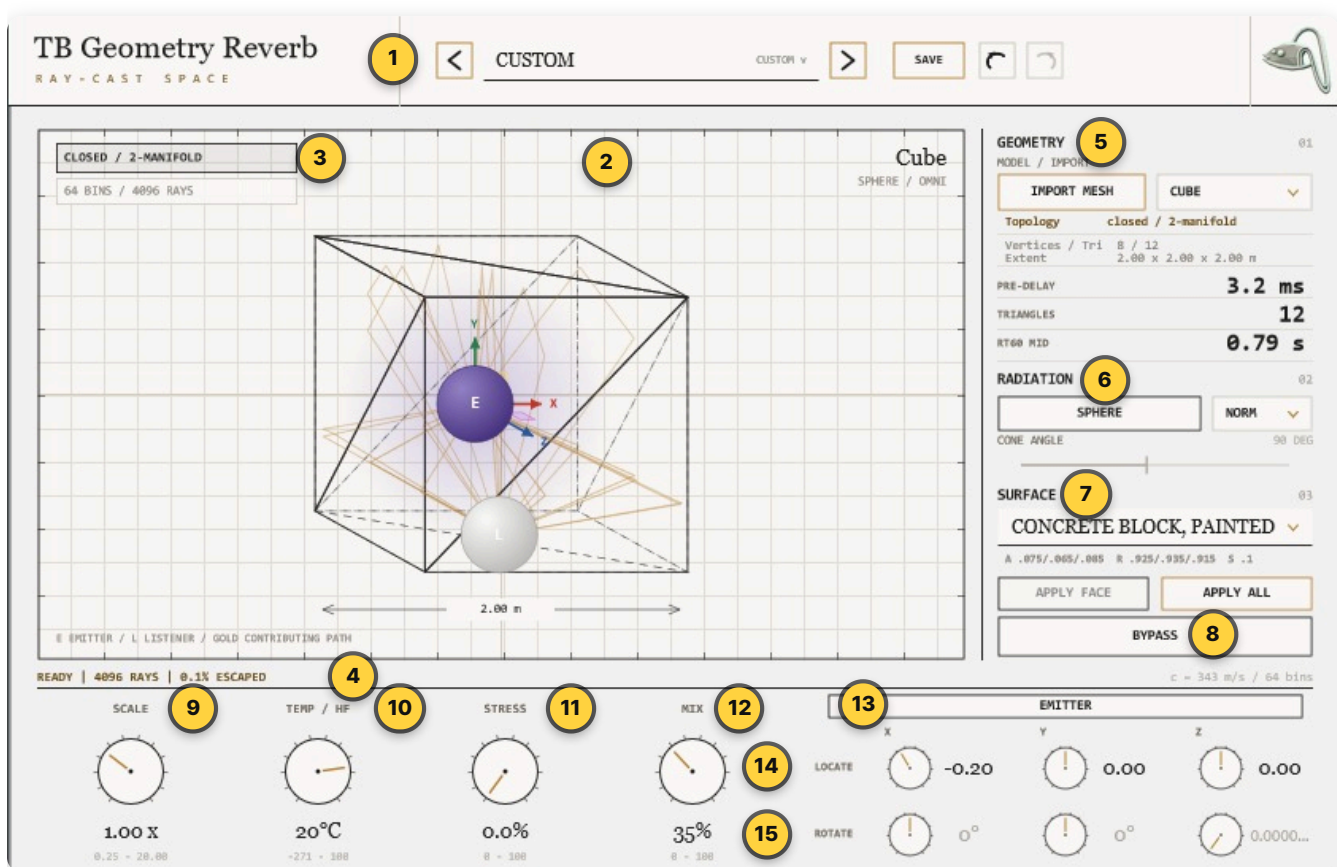
About this manual

This manual describes TB Geometry Reverb 1.0.0 for Windows (VST3). There is no macOS build at the time of writing.

02 Quick start

- 1 Insert TB Geometry Reverb on a track or an effects return. The default is a 2 m cube at SCALE 1.00×, with the emitter at X -0.20 and the listener at X 0.20, TEMP 20 °C, and MIX 35%.
- 2 Choose a starting point from the preset bar 1, or pick a model in GEOMETRY 5.
- 3 Set the size with SCALE 9 and the balance with MIX 12.
- 4 Choose a material in SURFACE 7 and press APPLY ALL.
- 5 Switch between EMITTER and LISTENER 13 and place them with LOCATE 14 and ROTATE 15. Wait until the status line 4 reads READY.

03 Interface



TB Geometry Reverb 1.0 interface with the default cube.

- 1 **Preset bar** — previous and next preset, the preset name and list, SAVE, Undo, and Redo.
- 2 **3D view** — the mesh, the emitter (E), the listener (L), and the contributing ray paths in gold.
- 3 **Mesh and ray info** — closed-manifold check, direction bins, and ray count.
- 4 **Status line** — READY, SIMULATING, BLOCKED, or an error, with the ray count and the share of escaped rays.
- 5 **GEOMETRY** — IMPORT MESH, the built-in models, mesh information, PRE-DELAY, TRIANGLES, and RT60 MID.
- 6 **RADIATION** — SPHERE or CONE, ray quality, and CONE ANGLE.
- 7 **SURFACE** — material list, coefficients, APPLY FACE, and APPLY ALL.
- 8 **BYPASS** — bypasses the reverb.
- 9 **SCALE** — size of the mesh.
- 10 **TEMP / HF** — air temperature.
- 11 **STRESS** — additional effect amount.
- 12 **MIX** — equal-power dry/wet.

13 EMITTER / LISTENER — switches the six knobs below between the emitter and the listener.

14 LOCATE — X, Y, and Z position of the selected point.

15 ROTATE — orientation of the selected point.

In the view, the selected point shows coloured X, Y, and Z handles. The scale bar under the model shows its size, and the line under the side panel shows the speed of sound (343 m/s at 20 °C).

04 Controls

Control	Range · default	What it does
SCALE	0.25 to 20× · 1.00×	Scales the mesh coordinates before path timing. Larger values make a bigger space with longer paths.
TEMP / HF	−271 to 100 °C · 20 °C	Air temperature. It sets the speed of sound and changes the high-frequency (HF) damping.
STRESS	0 to 100% · 0%	An additional effect amount that pushes the space harder; the DRIVE presets use it. Raise it gradually.
MIX	0 to 100% · 35%	Equal-power dry/wet mix.
RADIATION	SPHERE, CONE · SPHERE	SPHERE emits through all 64 direction bins; CONE focuses them into a cone.
CONE ANGLE	10 to 180° · 90°	Cone apex angle. Available only when RADIATION is CONE.
Ray quality	DRAFT, NORM, HIGH · NORM	How many rays are traced (4096 at NORM in the picture). Higher quality is more detailed and takes longer to calculate.
LOCATE X, Y, Z	−0.95 to 0.95 · emitter −0.20, 0, 0; listener 0.20, 0, 0	Position of the emitter or listener, relative to the mesh. Both must stay inside the closed mesh.
ROTATE	azimuth −180 to 180°, elevation −89 to 89°, roll −180 to 180° · 0°	Orientation of the emitter or listener. Listener angles are measured from the direction of the emitter, so 0° faces the source. Roll does nothing for the emitter, because a cone is rotationally symmetric.
BYPASS	Off, On · Off	Bypasses the reverb.

The following host parameters are stored in presets and the project but do not appear on the panel shown above.

Host parameter	Range · default	What it does
Rotation	−180 to 180° · 0°	Rotates the mesh.
Spin X, Spin Y, Spin Z	−6 to 6 rpm · 0	Rotation speed of the mesh around each axis.
Output	−18 to +12 dB · 0 dB	Output trim.
Super Grain	0 to 100% · 0%	An additional grain effect amount, used in several factory presets.

05 Geometry and materials

Models and import

- Built-in models: Tetrahedron, Cube, Sphere, Torus, Twisted Torus, Trefoil Knot, Superellipsoid, Harmonic Shell, Spiked Shell, Hyperboloid, Corrugated Drum, Capsule, and Auditorium.
- IMPORT MESH loads a Wavefront OBJ or FBX file in the background. Files must be 1 byte to 64 MiB, with up to 250,000 vertices and 500,000 triangles.
- Only a closed manifold mesh can produce a new acoustic snapshot. Open or non-manifold meshes are shown but not simulated, and self-intersections are not tested.
- The emitter and the listener must both stay inside the rotated closed mesh; otherwise the status line shows BLOCKED.
- PRE-DELAY, TRIANGLES, and RT60 MID are calculated from the current snapshot.

Materials

- Factory materials include Concrete, Concrete Block (Painted), Wood, Glass, Metal, and Curtain.
- The readout under the material shows A (absorption at low, mid, and high frequencies), R (reflection), and S (scattering).
- APPLY FACE assigns the material to the selected face; APPLY ALL assigns it to every surface. A mesh can use up to 256 material snapshots.
- The material menu also offers IMPORT JSON, LIBRARY, OPEN USER FOLDER, and RESCAN. User materials are .tbmaterial.json files with absorption and scattering coefficients from 0 to 1; up to 255 user material files are scanned.

06 Presets

Step through presets with the arrows, or open the list to choose a factory or user preset. SAVE stores the current settings, including the mesh and its materials, as a .tbgeometry file. Undo and Redo step through recent changes. In the table, TEMP is the air temperature and Quality the ray quality.

Preset	Model	SCALE	TEMP	Quality	MIX	Notes
START						
Cube Studio	Cube	1×	10 °C	NORM	35%	—
Tetra Chamber	Tetrahedron	1.5×	0 °C	NORM	40%	—
Sphere Studio	Sphere	2×	25 °C	NORM	42%	—
SPACE						
Close Chamber	Cube	0.45×	35 °C	NORM	30%	—
Concrete Hall	Cube	8×	−60 °C	HIGH	38%	—
Warm Room	Cube	3×	55 °C	NORM	34%	—
Large Vault	Sphere	4×	−120 °C	HIGH	48%	—
BEAM						
Focused Cone	Cube	3.5×	15 °C	HIGH	46%	CONE 35°

Preset	Model	SCALE	TEMP	Quality	MIX	Notes
Wide Cone	Tetrahedron	4×	5 °C	NORM	45%	CONE 125°
MATH						
Torus Ring	Torus	3×	12 °C	NORM	44%	—
Twisted Ring	Twisted Torus	3×	30 °C	HIGH	44%	—
Trefoil Knot	Trefoil Knot	2.5×	−40 °C	HIGH	50%	—
Squircle Room	Superellipsoid	6×	8 °C	NORM	40%	—
Harmonic Shell	Harmonic Shell	5×	−20 °C	HIGH	45%	—
Spiked Diffuser	Spiked Shell	4×	25 °C	HIGH	46%	—
Hyperboloid Waist	Hyperboloid	4.5×	0 °C	NORM	40%	—
Corrugated Drum	Corrugated Drum	3.5×	40 °C	HIGH	42%	—
Capsule Corridor	Capsule	4×	20 °C	NORM	38%	—
HALL						
Auditorium	Auditorium	1×	−80 °C	HIGH	34%	—
Auditorium Rear	Auditorium	1×	−40 °C	HIGH	40%	Super Grain 20%, Listener azimuth 155°
COMB						
Sphere Focus	Sphere	3×	0 °C	HIGH	50%	—
Golden Ratio	Cube	4×	10 °C	HIGH	45%	—
Axial Stack	Cube	3×	25 °C	NORM	46%	—
Corner Pair	Cube	5×	−30 °C	HIGH	44%	—
Squircle Null	Superellipsoid	4×	5 °C	HIGH	44%	—
Drum Ripple	Corrugated Drum	3.5×	20 °C	HIGH	45%	—
DRIVE						
Pushed Room	Cube	3×	30 °C	NORM	48%	STRESS 72%
Knot Grain	Trefoil Knot	2.5×	−60 °C	HIGH	52%	STRESS 35%, Super Grain 85%
Turned Away	Harmonic Shell	5×	15 °C	HIGH	46%	Super Grain 30%, Listener azimuth 95°, elevation 20°
Deep Freeze	Hyperboloid	6×	−265 °C	HIGH	50%	CONE 60°, STRESS 20%, Super Grain 45%

07 Recipes

Starting points for common jobs. Adjust them by ear.

Small studio

A short, natural room for vocals and instruments.

Preset Cube Studio

MIX 25 to 40%

- 1 Raise SCALE slightly if the room feels too tight.

Concrete hall

A large, hard space for drums or cinematic hits.

Preset Concrete Hall

SCALE 6 to 10×

- 1 Use an absorbent material such as Curtain if the tail gets too bright.

Directed source

A beam of sound aimed into the space.

Preset Focused Cone

CONE ANGLE 30 to 60°

- 1 Turn the emitter with ROTATE to aim the beam.

Abstract space

Unusual resonances from mathematical shapes.

Preset Trefoil Knot or Spiked Diffuser

Quality HIGH

- 1 Use DRAFT while editing and HIGH for the final sound.

08 Tips and troubleshooting

Symptom	What to check
The status line shows BLOCKED .	The emitter or listener is outside the closed mesh. Move LOCATE toward the centre, or check the mesh rotation.
An imported mesh does not sound.	It is not a closed manifold, is too large, or is not an OBJ or FBX file. Check the status line and the topology readout.
The tail is too long.	Lower SCALE, choose a more absorbent material, or lower MIX.
The tail is too bright or dull.	Change the surface material or TEMP / HF.
Updates take a long time.	Use DRAFT ray quality while editing.
Nothing changes.	MIX is at 0% or BYPASS is on.

09 Parameter reference

Panel label	Host parameter	Range	Default
SCALE	Scale	0.25 to 20×	1.00×
RADIATION	Radiation	Sphere, Cone	Sphere
CONE ANGLE	Cone Angle	10 to 180°	90°

Panel label	Host parameter	Range	Default
—	Rotation	−180 to 180°	0°
—	Spin Y	−6 to 6 rpm	0
TEMP / HF	Temperature	−271 to 100 °C	20 °C
Ray quality	Quality	Draft, Normal, High	Normal
LOCATE (emitter)	Source X, Y, Z	−0.95 to 0.95	−0.20, 0, 0
LOCATE (listener)	Receiver X, Y, Z	−0.95 to 0.95	0.20, 0, 0
MIX	Mix	0 to 100%	35%
—	Output	−18 to +12 dB	0 dB
BYPASS	Bypass	Off, On	Off
—	Spin X, Spin Z	−6 to 6 rpm	0
ROTATE (emitter)	Emitter Azimuth, Emitter Elevation	−180 to 180°, −89 to 89°	0°
STRESS	Stress	0 to 100%	0%
ROTATE (listener)	Listener Azimuth, Elevation, Roll	−180 to 180°, −89 to 89°, −180 to 180°	0°
—	Super Grain	0 to 100%	0%

10 Version history

Version	Changes
1.0.0 · 2026-08-17	First release: a geometry-based reverb with mesh import, closed-mesh validation, and emitter and listener position checks; a 64-direction snapshot ray model with three ray-quality levels and an eight-line FDN pass; material, scale, and radiation-pattern control; state, preset, and host-state handling for snapshot calculation.

11 Support and license

Report problems or request features at github.com/TaebeumKim/TB_Audio-Plugins/issues. Include the plug-in version, your DAW, and your operating system.

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